Akash Kumar Singh

Full Stack Developer | React, Node.js, Express, PostgreSQL

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PROFESSIONAL SUMMARY

Full stack developer and MCA (AI & ML) student who builds and ships web applications end to end, from database schema and REST APIs to React interfaces. Built and delivered an internal billing and invoicing system for a travel company as a solo freelancer, and a real-time collaborative app that syncs a one-million-checkbox grid across users with WebSockets and Redis. Four years in a finance team gives me a working understanding of how admin and operations staff actually use internal tools, which is directly useful for ERP-style products.

TECHNICAL SKILLS

- **Frontend:** React, Next.js, JavaScript (ES6+), HTML5, CSS3, responsive UI
- **Backend:** Node.js, Express.js, REST API design, WebSockets, FastAPI, Python
- **Databases:** PostgreSQL, Prisma ORM, MySQL, MongoDB, SQLite, Redis (caching, bitfields, Pub/Sub)
- **Auth & Integrations:** OAuth 2.0 (Google, GitHub), Firebase Authentication, JWT, Twilio/WhatsApp APIs, Anthropic Claude API
- **Tools:** Git, GitHub, Linux, Vercel, Netlify, pytest
- **Practices:** Role-based internal tools, real-time sync, lazy loading and pagination, offline-first UI, requirement gathering

EXPERIENCE

Freelance Full Stack Developer | Self-employed (client projects alongside studies)

- Designed and built a staff-only billing and invoice web app for a tour and travel company using React, Node.js/Express, PostgreSQL with Prisma, and Redis.
- Owned the full lifecycle on my own: requirements with the client, database design, API development, frontend, and deployment.
- Added AI-assisted features that generate trip summaries, flag unusual charges, and suggest line items, cutting manual checking for staff.

Accountant | Phoenix Engineers | Sep 2022 – Present

- Reconcile bank statements against Tally Prime ledgers and categorize transactions by counterparty, giving me first-hand experience as a daily user of business software.
- Work with management on finance processes and month-end reporting, which shaped how I design admin and back-office screens.

PROJECTS

CheckGrid – Real-Time Collaborative Checkbox Grid | Node.js, Express, WebSockets, Redis, OAuth 2.0

- Built a web app where many users toggle a shared grid of one million checkboxes and see each other's changes live.
- Stored grid state in a Redis bitfield to keep a million booleans compact in memory, and used Redis Pub/Sub to broadcast updates across server instances.
- Added Google and GitHub sign-in with OAuth 2.0 and designed the UI for the app, landing page, and auth flow.

Shopfloor Sentinel – Factory Security Dashboard (TCS iON Hackathon) | React, Node.js, Python, Claude API | Oct 2026 – Nov 2026

- Built a monitoring dashboard backed by a Python gateway (26 tests) that detects port scans, RDP brute force, Modbus writes, and USB insertion on factory networks.
- Set up alert escalation over WhatsApp and SMS with a retry queue, SMS fallback, and scheduled daily summaries for owners.
- Added an AI triage step that writes plain-language incident summaries for non-technical users.

SENTINEL – LLM Security Platform | Python, Next.js, Claude API | Jun 2026

- Built a Next.js dashboard (12 components, 6 API endpoints) on top of a Python library with 17 pytest tests.
- Cut average prompt-injection attack success from about 77% to 11% in a benchmark of around 100 live API calls.

EDUCATION

- **Master of Computer Applications (MCA), Artificial Intelligence & Machine Learning** | Amity University Online | Jan 2026 – Present
- **Master of Business Administration (MBA), Finance** | IGNOU | Jan 2022 – Dec 2025
- **Bachelor of Commerce (B.Com), Accountancy** | Karim City College | Aug 2013 – Jul 2016 | 63.88%

CERTIFICATIONS & ACHIEVEMENTS

- Full Stack Web Development Certification | Physics Wallah | Aug 2023 – Feb 2025
- Ranked 15th of 133 participants, Chai Forms Hackathon, ChaiCode | May 2026